

Introduction to Binary Numbers and Hexadecimals

Prof. Marios S. Pattichis, FEAMBES

Image and Video Processing and Communications Lab (ivPCL)
Dept. of ECE, University of New Mexico, USA
www.ivpcl.org

Outline

Intro to binary
and
hexadecimals

M.S. Pattichis

The Decimals (1 of 2)

Counting using base 10:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

To avoid confusion when counting base 10, we add a subscript:

$0_{10}, 1_{10}, 2_{10}, 3_{10}, 4_{10}, 5_{10}, 6_{10}, 7_{10}, 8_{10}, 9_{10}, 10_{10}, 11_{10}.$

We recall that:

$$10_{10} = 1 \cdot 10 + 0 \cdot 1.$$

Also, note that:

$$11_{10} = 1 \cdot 10 + 1 \cdot 1.$$

The Decimals (2 of 2)

For higher powers, we use powers of 10:

$$10^3, 10^2, 10^1 = 10, 10^0 = 1.$$

Thus, we have the following representation:

$$1234_{10} = 1 \cdot 10^3 + 2 \cdot 10^2 + 3 \cdot 10 + 4 \cdot 1.$$

Binary Numbers (1 of 2)

For binary numbers, we use powers of 2:

$$2^3 = 8, 2^2 = 4, 2^1 = 2, 2^0 = 1.$$

For binary numbers, we only use 0 and 1 for the digits.
Here are the first 5 numbers:

$$0_2 = 0 \cdot 2^0 = 0 \cdot 1 = 0_{10}$$

$$1_2 = 1 \cdot 1 = 1_{10}$$

$$10_2 = 1 \cdot 2^1 + 0 \cdot 1 = 2_{10}$$

$$11_2 = 1 \cdot 2^1 + 1 \cdot 1 = 3_{10}$$

$$100_2 = 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 1 = 4_{10}$$

Binary Numbers (2 of 2)

Intro to binary
and
hexadecimals

M.S. Pattichis

Fill in the following numbers:

$$101_2 = 5_{10} =$$

$$110_2 = 6_{10} =$$

$$111_2 = 7_{10} =$$

$$1000_2 = 8_{10} =$$

$$1001_2 = 9_{10} =$$

Hexadecimals (1 of 3)

For hexadecimals, we use powers of 16:

$$16^1 = 16, 16^0 = 1.$$

For 0 to 9, we use the same digits as for the decimals!

For 10 to 15, we switch to letters:

$$A_{16} = 10_{10} = 10.$$

$$B_{16} = 11_{10} = 11.$$

$$C_{16} = 12_{10} = 12.$$

$$D_{16} = 13_{10} = 13.$$

$$E_{16} = 14_{10} = 14.$$

$$F_{16} = 15_{10} = 15.$$

Hexadecimals (2 of 3)

When using A to F, we assume that we are in hexadecimals.

Like before, we now use the powers:

$$\begin{aligned}AB &= A \cdot 16^1 + B \cdot 16^0 \\&= 10 \cdot 16 + 11 \cdot 1 \\&= 160 + 11 \\&= 171\end{aligned}$$

Hexadecimals (3 of 3)

Intro to binary
and
hexadecimals

M.S. Pattichis

You can break hexadecimal numbers into groups of 4 to get the binary representation:

$$AB = 10101011$$

What are the binary representations for AA, FF?